

# Stenting as a treatment for stroke

MATH 213

# Group 1: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.

1. What type of study was this? (Observational study or experiment?) How can you tell?

Experiment because they're actively controlling the outcome, like assigning people into groups, and not just observing the outcome

2. Is there the potential to be able to draw causal conclusions, given the type of study?

Yes, because it's an experiment, meaning that other variables can be mitigated and controlled.

A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

Explanatory: The treatment given (categorical)

Response: Frequency of strokes (numerical)

3. What are the cases in this study?

The 451 at risk patients

In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

| patient   | group     | Number of strokes at 30 days | Number of strokes at 365 |
|-----------|-----------|------------------------------|--------------------------|
| Patient 1 | treatment | 0                            | 1                        |
| Patient 2 | control   | 7                            | 9                        |
| Patient 3 | treatment | 1                            | 2                        |
| Patient 4 | control   | 4                            | 6                        |

**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**

Support your conclusion with some specific numbers.  
Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups? What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

Stents don't seem to work, because for both segments of days, the control groups both had more people without events. Meanwhile, the treatment groups consistently had more strokes over the control groups within both time frames. It might be hard to fully say if this is accurate since there are more people in the control group than the treatment, but it can't be enough to affect the numbers like this with only 3 people more with a 33-13 ratio. In a year 62% of the stroke ratio were in the treatment group while only 38% came from the control group

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

Only 12% of the control group had a stroke. Meanwhile, 20% of the control group had a stroke within the same time span.

## Group 2: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.

**1. What type of study was this? (Observational study or experiment?) How can you tell?**

This was an experiment because they controlled the explanatory variable (their type of treatment).

**2. Is there the potential to be able to draw causal conclusions, given the type of study?**

Yes, there is potential to draw causal conclusions because it is an experiment and they randomly assigned patients to the control and treatment groups. My only concern is whether there may be possible bias in the sample selection. It does not go into detail in how they selected the sample, however if there is no bias then there is potential to draw causal conclusions.



A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

### 3. What are the cases in this study?

The cases in this study are the 451 at-risk patients who volunteered for the experiment.

### 4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

The explanatory variable is whether or not they received a stent. The response variable is whether or not they had a stroke either within 30 days, or 365 days after receiving the treatment.

In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

| Patients  | Group     | Stroke within 30 Days? | Stroke within 356 days? |
|-----------|-----------|------------------------|-------------------------|
| Patient 1 | Treatment | No                     | Yes                     |
| Patient 2 | Control   | No                     | No                      |

**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**

Support your conclusion with some specific numbers.

Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups?*

*What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

The data suggests that the stents made it more likely for a person to have a stroke. For example, during the first 30 days, the control group only had 13 strokes while the treatment group had 33 strokes. Then within 0-365 days, the control group only had 28 strokes while the treatment group had 45 strokes. This data suggests that stenting can increase the chance of stroke.

|                   |                 | Treatment | Control |
|-------------------|-----------------|-----------|---------|
| <b>0-30 Days</b>  | <b>Stroke</b>   | 33        | 13      |
|                   | <b>No event</b> | 191       | 214     |
| <b>0-365 Days</b> | <b>Stroke</b>   | 45        | 28      |
|                   | <b>No event</b> | 179       | 199     |

White board questions

- 1. Of the people who had a stroke in a year, what proportion were in the treatment group?**

About 62% of people who had a stroke in a year were in the treatment group.  $45/73$

- 2. Of the people in the treatment group, what proportion had a stroke in the first year?**

About 20% of people in the treatment group had a stroke in the first year.  $45/224$

## Group 3: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.

1. What type of study was this? (Observational study or experiment?) How can you tell?  
Experiment - because there is a control on the explanatory variable (treatment vs. control group)

2. Is there the potential to be able to draw causal conclusions, given the type of study?

Yes

A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

3. What are the cases in this study?

At-risk patients

4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

Explanatory - Treatment or control group (categorical)

Response - Stroke or no event at 30 days and 365 days(categorical)

In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

| #         | Group     | Stroke or no event(30 days) | Stroke or no event (365 days) |
|-----------|-----------|-----------------------------|-------------------------------|
| Patient 1 | Treatment | No event                    | Stroke                        |
| Patient 2 | Control   | Stroke                      | Stroke                        |

**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**

Support your conclusion with some specific numbers.

Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups?*

*What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

Based on the data, stenting is not effective for preventing strokes – in fact, it may even increase the likelihood of having a stroke. At both time points (30-day and 365-day), there were more participants in the treatment group who had a stroke than the control group. By the end of the 365 days, 28 control group participants had a stroke which is less than the 33 treatment group participants who had a stroke within the first 30 days.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

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# Group 4: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.

1. What type of study was this? (Observational study or experiment?) How can you tell?

The type of study is experiment. This is because the researchers are putting people into to different groups, and changing whether or not one group is given stents and one is not.

2. Is there the potential to be able to draw causal conclusions, given the type of study?

Yes, there is a potential to be able to draw causal conclusions because there is random assignment and comparison of groups.

A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

Explanatory: Whether a patient received a stent (Categorical)

Response: Occurance of a stroke (Categorical)

3. What are the cases in this study?

The 451 at risk patients

In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

|           |           |                       |                        |
|-----------|-----------|-----------------------|------------------------|
| Patients  | group     | Stroke within 30 days | Stroke within 365 days |
| Patient 1 | treatment | yes                   | yes                    |
| Patient 2 | control   | no                    | no                     |

**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**

Support your conclusion with some specific numbers.

Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups?*

*What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

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## Group 5: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.

1. What type of study was this? (Observational study or experiment?) How can you tell?
2. Is there the potential to be able to draw causal conclusions, given the type of study?

A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

3. What are the cases in this study?



In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

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**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**

Support your conclusion with some specific numbers.

Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups?*

*What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

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## Group 6: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.

1. What type of study was this? (Observational study or experiment?) How can you tell?
2. Is there the potential to be able to draw causal conclusions, given the type of study?

A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

3. What are the cases in this study?

In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

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**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**  
Support your conclusion with some specific numbers.  
Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups? What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

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## Group 7: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.



1. What type of study was this? (Observational study or experiment?) How can you tell?

**Experiment, the researchers assigned the control groups.**

2. Is there the potential to be able to draw causal conclusions, given the type of study?

Yes, there is a clear treatment group and a clear control group

A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

Explanatory: Taking stents

Response: Having a stroke.

3. What are the cases in this study?

Patients with high risk of stroke.

In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

| Patient   | Treatment Group | Did they suffer from a stroke within 30 days of enrollment? | Did they suffer from a stroke within a year of enrollment? |
|-----------|-----------------|-------------------------------------------------------------|------------------------------------------------------------|
| Patient 1 | stent           | yes                                                         | yes                                                        |
| Patient 2 | No stent        | no                                                          | yes                                                        |

**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**  
Support your conclusion with some specific numbers.  
Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups? What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

Stenting does not seem to help prevent strokes from happening. The data actually suggests that it makes people at risk, more likely to have a stroke than those that do not use stents.

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## Group 8: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.

1. What type of study was this? (Observational study or experiment?) How can you tell?
2. Is there the potential to be able to draw causal conclusions, given the type of study?

A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

3. What are the cases in this study?

In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

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**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**  
Support your conclusion with some specific numbers.  
Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups? What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

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# Group 9: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.

1. What type of study was this? (Observational study or experiment?) How can you tell?
2. Is there the potential to be able to draw causal conclusions, given the type of study?

A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

3. What are the cases in this study?

In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

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**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**  
Support your conclusion with some specific numbers.  
Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups? What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

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# Group 10: Introduction

We will consider an experiment that studies effectiveness of **stents** in treating patients at risk of stroke.

Stents are devices put inside blood vessels that assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. **Does the use of stents reduce the risk of stroke?**

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group:** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group:** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group.

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment.

1. What type of study was this? (Observational study or experiment?) How can you tell?

Experiment - they were able to control who received the stents and who did not.  
Also, it says in the first line that it was an experiment...

2. Is there the potential to be able to draw causal conclusions, given the type of study?

Yes, there is potential to draw causal conclusions because the study is an experiment. In addition, random assignment was used, which also supports the potential to draw causal conclusions.



A **contingency table** (a type of **frequency table**) is a way to summarize the relationships between two (or more) categorical variables. The following contingency table summarizes the results of this study.

|            |          | Treatment | Control |
|------------|----------|-----------|---------|
| 0-30 Days  | Stroke   | 33        | 13      |
|            | No event | 191       | 214     |
| 0-365 Days | Stroke   | 45        | 28      |
|            | No event | 179       | 199     |

4. What are the main explanatory and response variables for this study? Classify each as numerical or categorical.

Did or did not have stroke - response; categorical

Grouping - explanatory; category

3. What are the cases in this study?

451 at-risk patients

In the table below, make the first few rows of a hypothetical data frame for this study. Fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. For Patient 1, suppose they were stented and had a stroke on day 100 but no earlier than that. For Patient 2, make up your own variable values.

| Cases | Group     | Stroke event (w/in 30 days) | Stroke event ( w/in 365) |
|-------|-----------|-----------------------------|--------------------------|
| P1    | Treatment | No event                    | Stroke                   |
| P2    | Control   | No event                    | No Event                 |

**Judging from the table, what do you make of the effectiveness of stenting for preventing strokes?**

Support your conclusion with some specific numbers.

Hint: *What makes it difficult to directly use the numbers that are printed in the table to compare the groups?*

*What can we calculate with these numbers to more easily compare them, do you think? (Do these calculations and use them as part of your explanation.)*

|                   |                 | Treatment | Control |
|-------------------|-----------------|-----------|---------|
| <b>0-30 Days</b>  | <b>Stroke</b>   | 33        | 13      |
|                   | <b>No event</b> | 191       | 214     |
| <b>0-365 Days</b> | <b>Stroke</b>   | 45        | 28      |
|                   | <b>No event</b> | 179       | 199     |

Another textbox here:

There is no way to clearly determine effectiveness as there were more in the treatment group in the first 30 days to have strokes but for the remainder of the year more control group patients had strokes than those in the treatment group. If the stents were effective would they have had an immediate effect? Or would there have been a more significant difference over the course of the entire year? The numbers don't give a wide enough margin to give clear answers.

**Class answer: Seems ineffective from data because over both shorter term and longer term, no clear, distinguishable improvement; even possible evidence of harm**